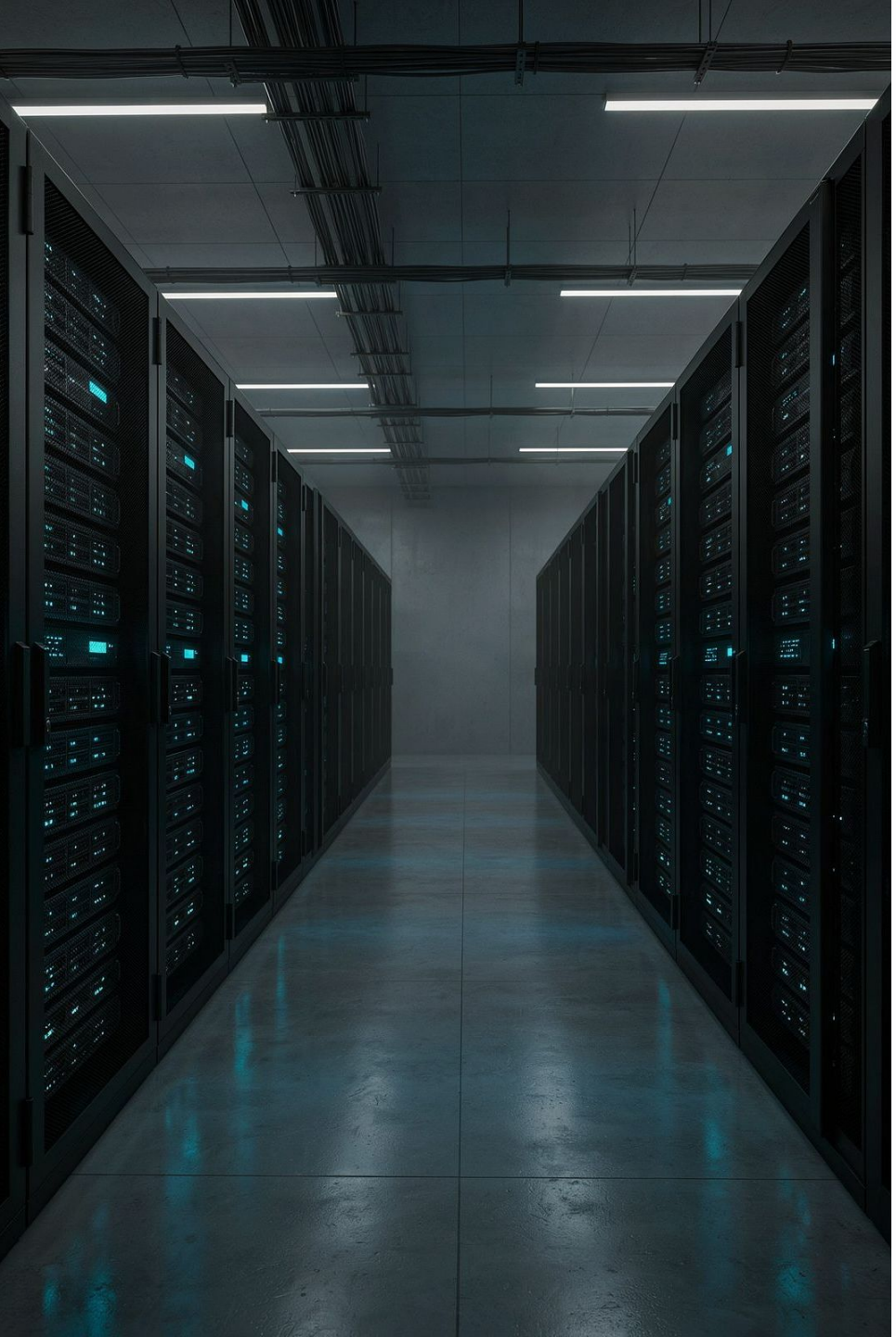


Building on AWS AI: Bedrock, SageMaker & Landing Zones

A practical deep-dive into Amazon's AI infrastructure — from foundation models to production-ready workloads. Learn how real organizations are designing, deploying, and governing AI at scale on AWS.

AWS AI PLATFORM

2024 DEEP DIVE

Today's Agenda

What We'll Cover

Three powerful pillars of the AWS AI platform — explored through real-world scenarios, practical architecture, and hands-on context so you can map these services to your own organization's needs.



Amazon Bedrock

Foundation models as a service — how to use Claude, Titan, Llama and others without managing infrastructure



AI Landing Zone Design

Governance, networking, and security blueprints for running AI workloads safely at enterprise scale



SageMaker AI Studio

The unified IDE for data scientists and ML engineers — from notebook to production model

CHAPTER 1

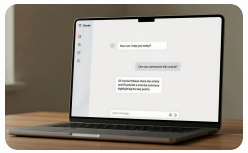
Amazon Bedrock — AI Without the Heavy Lifting

Think of Amazon Bedrock as a well-stocked toolbox of AI brains — you pick the model that fits your task, plug it in via API, and start building. No GPUs to manage, no model training from scratch.



What Is Amazon Bedrock?

Amazon Bedrock is a fully managed service that gives developers access to high-quality foundation models (FMs) from Amazon and leading AI companies — all through a single API. Think of it like **Netflix for AI models**: one subscription, many options, no infrastructure headaches.



Anthropic Claude

Best for long-document summarization, complex reasoning, and customer-facing chat



Amazon Titan

AWS-native models optimized for text generation, embeddings, and image synthesis



Meta Llama

Open-weight models ideal for custom fine-tuning and cost-sensitive use cases



Stability AI

Image generation models for creative content, product mockups, and design workflows

Bedrock in Action — Key Capabilities

Bedrock goes beyond raw model access. It provides a full layer of enterprise-grade features to build **safe, scalable, production-ready** AI applications — without reinventing the wheel.

Retrieval-Augmented Generation (RAG)

Connect your company's internal documents — contracts, wikis, manuals — to a foundation model. The model answers questions using *your* data, not just its training data. Real example: a legal firm using RAG to query 10,000 past case documents instantly.

Agents for Bedrock

Build AI agents that can take multi-step actions — like searching a database, calling an API, and writing an email — autonomously. Real example: an HR bot that pulls employee data, checks policy documents, and drafts an onboarding plan in one shot.

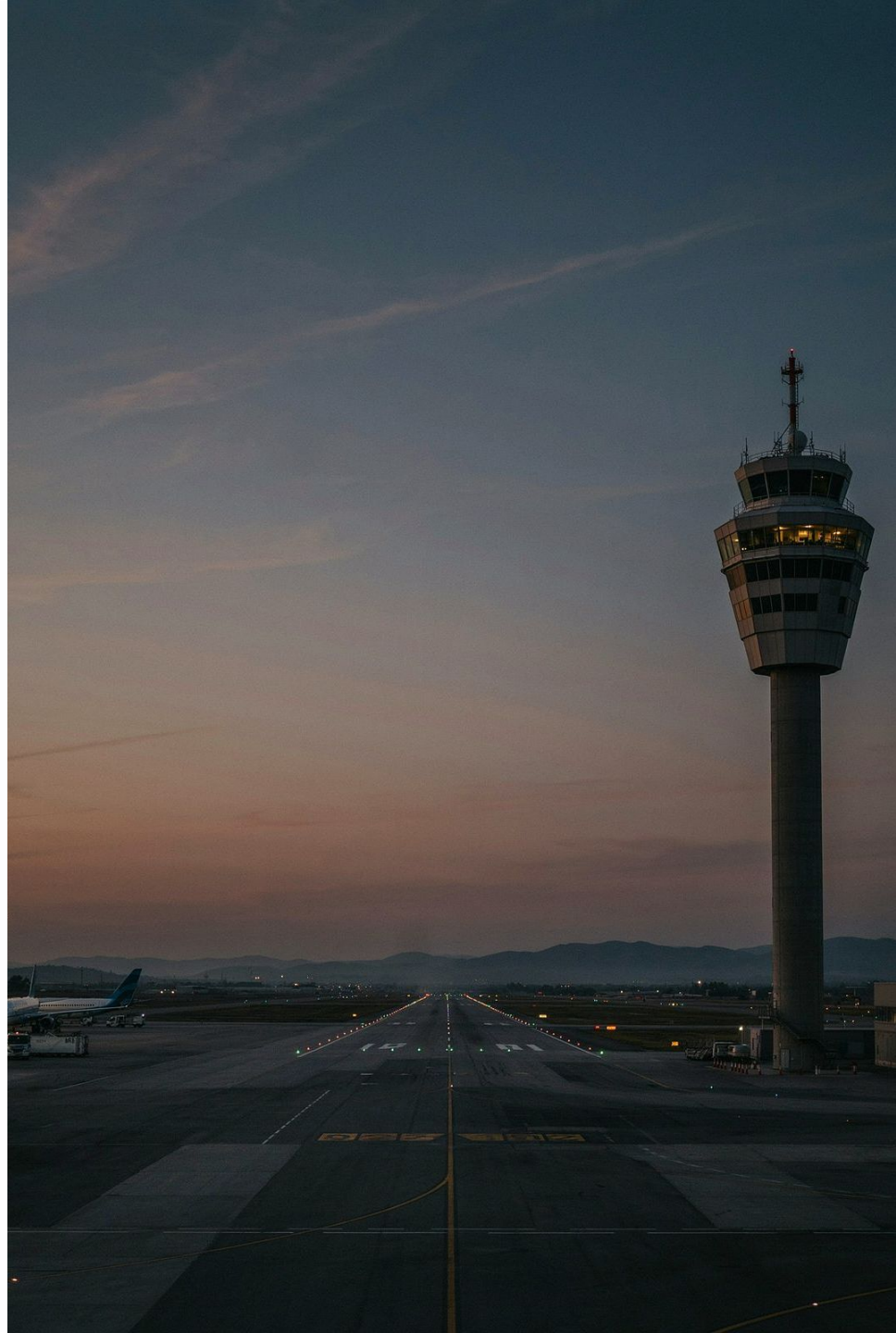
Model Evaluation & Guardrails

Built-in tools to test model accuracy and enforce content safety policies — blocking harmful outputs before they reach end users. Essential for regulated industries like finance and healthcare.

Chapter 2

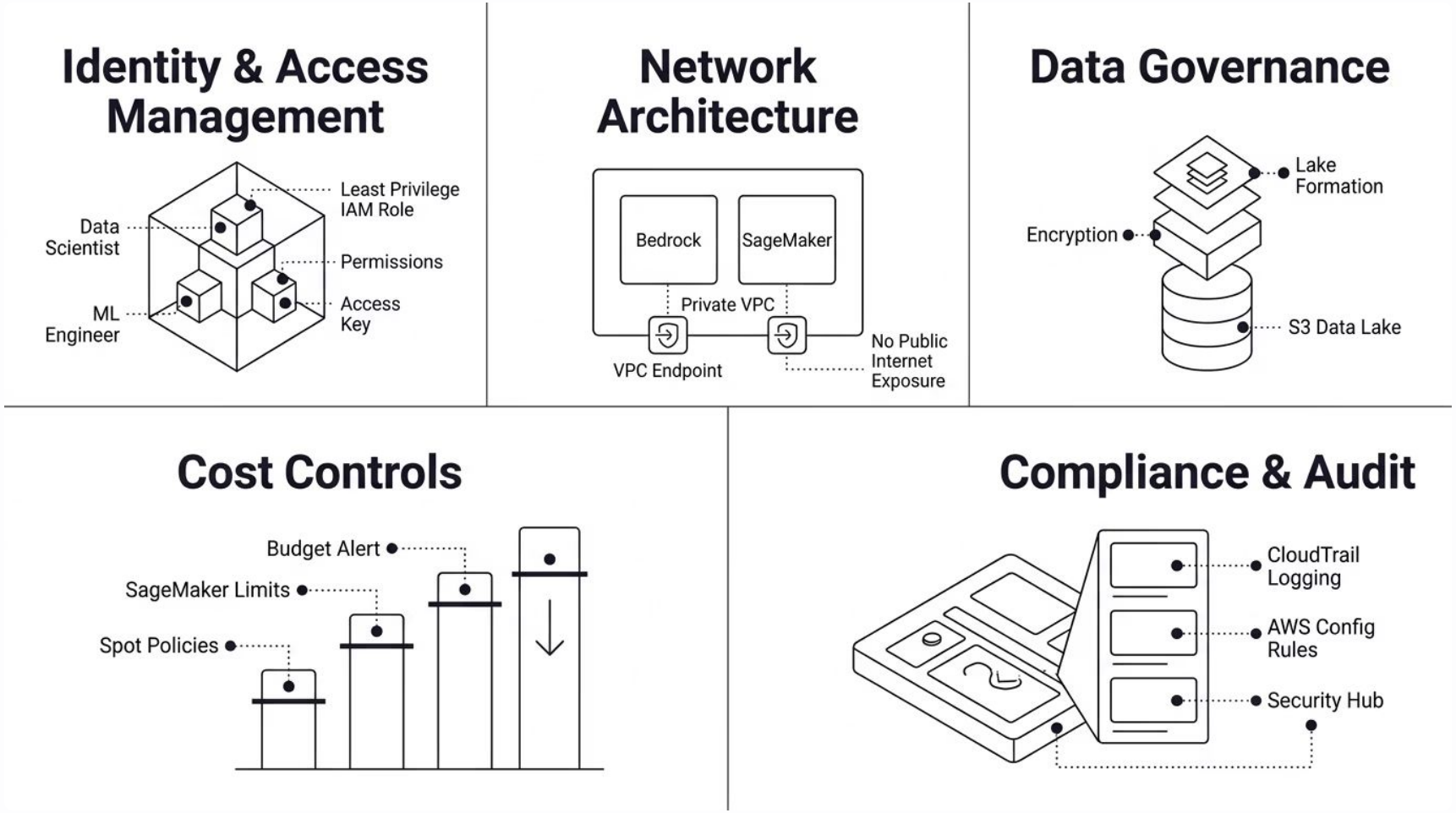
AI Workload Landing Zone — Building a Safe Runway

Just as an airport needs a safe, well-organized runway before planes can land, your AWS environment needs a structured "landing zone" before AI workloads can operate safely at scale.



Designing Your AI Landing Zone

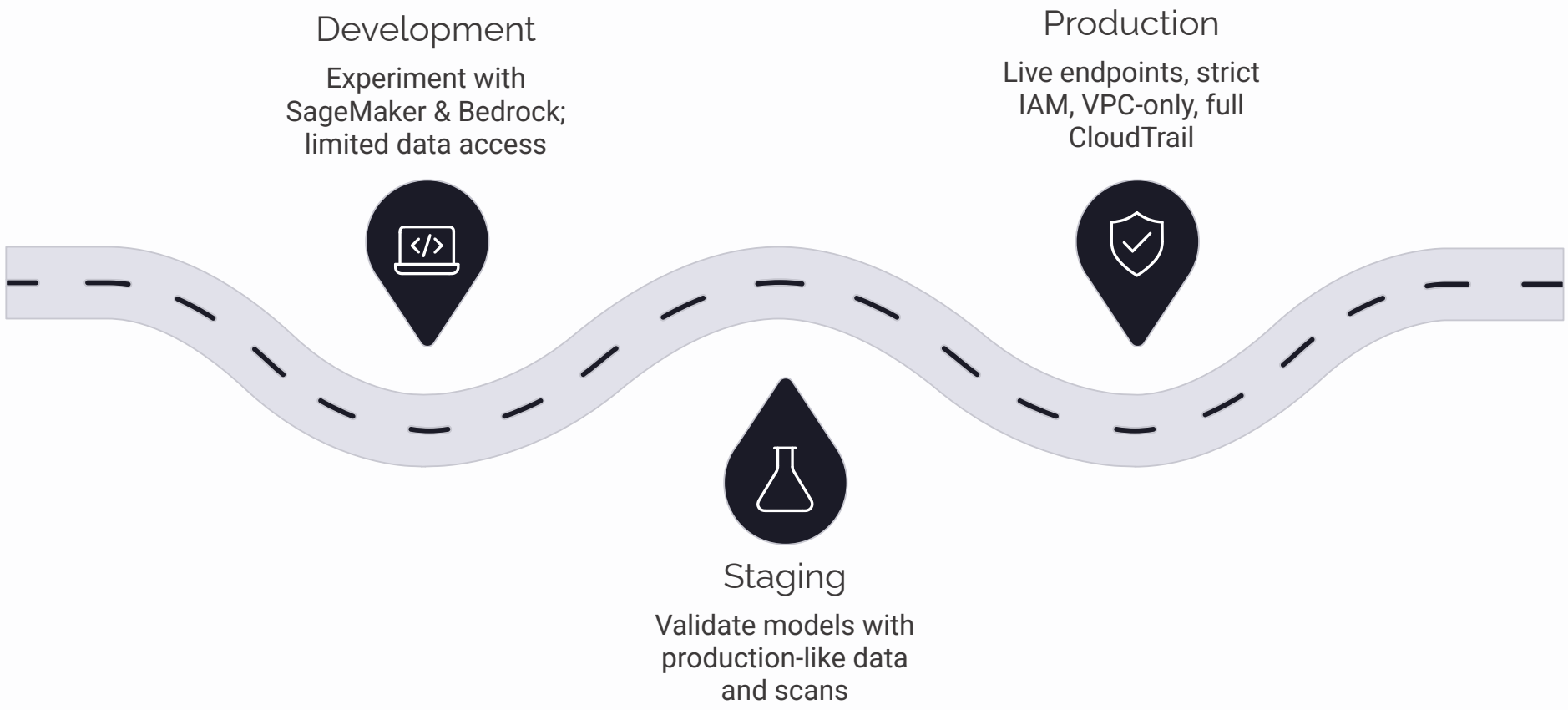
An AI Landing Zone is a **pre-configured, governed AWS environment** purpose-built for AI/ML workloads. It applies AWS Control Tower principles with AI-specific guardrails — handling security, networking, identity, and cost controls from day one.



Each pillar maps to a real organizational concern — from a data scientist accidentally accessing production data, to an ML job running unchecked and generating a $50,000 bill overnight.

Real-World Landing Zone Architecture

Here's how a mid-sized financial services company structured their AI landing zone across three AWS accounts — a common pattern called the **three-account model**.



Separating environments prevents a rogue experiment from affecting live customer-facing AI models — the same principle banks use to separate trading desks from back-office systems.

CHAPTER 3

Amazon SageMaker AI Studio — The ML Engineer's Cockpit

SageMaker AI Studio is AWS's fully integrated development environment for machine learning — a single web-based interface where data scientists, ML engineers, and analysts collaborate to build, train, and deploy models.



SageMaker AI Studio — Core Capabilities

SageMaker AI Studio removes the fragmented toolchain problem. Instead of juggling separate tools for notebooks, training jobs, experiments, and deployment, everything lives in **one unified IDE**. Here's what makes it powerful in practice:

📓 JupyterLab Notebooks

Fully managed, auto-scaling notebooks backed by on-demand compute. A data scientist can switch from a CPU instance to a GPU instance in seconds — no IT ticket required. Pay only for what you use.

🔬 Experiments & Model Registry

Track every training run — hyperparameters, metrics, artifacts — automatically. The Model Registry acts like version control for ML models, letting teams promote models from staging to production with an approval workflow.

🚀 One-Click Deployment

Deploy trained models to real-time endpoints, batch transform jobs, or serverless inference — directly from the Studio UI. Real example: a retail company deploying a demand-forecasting model in under 10 minutes, no DevOps team needed.

🤝 Collaboration & Domains

SageMaker Domains give each team or business unit its own isolated workspace — shared data access, but separate compute and permissions. Perfect for large enterprises with multiple ML teams working in parallel.

Putting It All Together

These three AWS pillars work best when combined. Here's how a real e-commerce company might use all three to build a product recommendation engine:



SageMaker Studio

Data scientists explore customer behavior data in notebooks, train a custom recommendation model, and register it in the Model Registry

AI Landing Zone

The model is promoted through Dev → Staging → Prod accounts with automated security checks, IAM controls, and cost guardrails at each gate

Amazon Bedrock

A Bedrock Agent combines the custom model with Claude to generate personalized product descriptions and handle natural language shopping queries in real time

Result: Faster time-to-market, lower operational risk, and a fully auditable AI pipeline — from raw data to live customer experience.

Key Takeaways & Next Steps

You now have a clear map of AWS's AI platform. The path forward is to start small, establish governance early, and build momentum with quick wins before scaling to enterprise-wide AI adoption.

🧠 Amazon Bedrock

Start with a RAG prototype using your internal documents. Use Bedrock Guardrails from day one to enforce content safety — even in experiments.

🏗️ Landing Zone

Deploy AWS Control Tower with the AI/ML customizations. Establish the three-account model before your first production workload — retrofitting governance is 10x harder.

🔬 SageMaker Studio

Create a SageMaker Domain for your ML team today. Enable experiment tracking from the first model run — you'll thank yourself when comparing 50 experiments three months later.

The best time to build the right AI foundation was six months ago. The second best time is today.